Continuous Curing Oven (IND-OVEN) Equipment Manual and SOP

VINDHYA PRECISION WORKS ## EQUIPMENT MANUAL AND MAINTENANCE STANDARD OPERATING PROCEDURE ### ASSET CLASS: Continuous Curing Oven (IND-OVEN) Document ID: SOP-IND-OVEN-014 Revision: 4.2 Effective Date: 14 January 2025 Owner: Maintenance Engineering Lead

1. GENERAL DESCRIPTION AND APPLICABILITY

1.1 This document governs all operation, routine maintenance, condition monitoring, and corrective intervention for asset class IND-OVEN (Continuous Curing Oven) installed across Vindhya Precision Works facilities.

1.2 Primary maintenance responsibility is assigned to the Electrical trade, with Instrumentation providing mandatory support for control loops, pyrometric profiling, and safety interlock verification.

1.3 Healthy equipment operating within baseline parameters shall exhibit vibration levels between 1.2 mm/s and 4.0 mm/s RMS, and external surface temperatures between $38\,^{\circ}\text{C}$ and $66\,^{\circ}\text{C}$.

1.4 This asset class operates at atmospheric pressure; therefore, no monitored process pressure limits, alarms, or trips apply to IND-OVEN.

2. HAZARDOUS ENERGY CONTROL AND ISOLATION

2.1 Prior to undertaking any corrective maintenance, mechanical adjustment, or internal inspection, technicians shall execute a complete energy isolation in accordance with Vindhya Precision Works Safety Standard VPW-LOTO-01.
2.2 The hazardous energy sources associated explicitly with asset class IND-OVEN are:
1. Electrical: 415V AC 3-phase supply via Main Isolator Circuit Breaker MCC-B14, and 24V DC control power.
2. Thermal: Internal heating elements and refractory lining retaining residual heat exceeding $250\,^{\circ}\text{C}$.

2.3 Thermal energy dissipation requires a mandatory cool-down period until internal ambient temperature drops below $40\,^{\circ}\text{C}$ before technicians may enter the curing chamber.

3. OPERATING RANGES, THRESHOLDS, AND ALARM SETPOINTS

3.1 The condition-monitoring system shall continuously poll asset class IND-OVEN against the following hard limits. Exceeding warning thresholds triggers a restricted alert to the control room; exceeding trip thresholds initiates an immediate, automated category-1 safety shutdown.

Parameter Monitored	Healthy Baseline	Warning Setpoint	Trip Setpoint	Action on Trip
Vibration (Drive End / Blower)	$1.2 - 4.0\text{ mm/s}$ RMS	4.8 mm/s RMS	7.5 mm/s RMS	Immediate 415V trip, coast to stop
Surface Temperature (Casing)	$38 - 66\,^{\circ}\text{C}$	$90\,^{\circ}\text{C}$	$105\,^{\circ}\text{C}$	Interlock element power cut, run exhaust fan
Current (Drive Motor)	$28.0 - 35.0\text{ A}$	39.5 A	44.0 A	Overload relay trip, alarm annunciated
Process Pressure	*None*	*None*	*None*	*Not applicable to IND-OVEN*

3.2 Any vibration reading remaining between 4.8 mm/s and 7.49 mm/s RMS for more than 120 continuous seconds shall escalate automatically to a maintenance work order generation.

4. PREVENTIVE MAINTENANCE SCHEDULE

4.1 The base preventive maintenance (PM) interval for asset class IND-OVEN is exactly 1500 running hours (± 25 hours tolerance).
4.2 Maintenance tasks shall be executed according to the following running-hour hierarchy:

* 1500 Running Hours (PM-1):
- Inspect and test electrical termination tightness inside the main control cabinet (torque to specification per Section 7).
- Lubricate conveyor drive bearings using high-temperature synthetic grease (maximum 15 strokes per fitting).
- Inspect condition of VPW-P-00038 (Family belt) and VPW-P-00039 (Relationship belt) for fraying, tension loss, or cracking; adjust alignment.
- Verify calibration of thermocouple arrays against secondary standard.
- Clean blower intake mesh and inspect VPW-P-00058 (Reason seal) for gas leakage or thermal hardening.

* 3000 Running Hours (PM-2):
- Execute all PM-1 tasks.
- Inspect and test power contactors VPW-P-00048 (Part contactor) and VPW-P-00052 (Material contactor) for contact pitting, arc erosion, and coil resistance.
- Inspect pneumatic/hydraulic actuation paths, replacing VPW-P-00004 (Forget valve) and VPW-P-00069 (Race valve) O-rings and seals.
- Check structural integrity of internal belt support rollers and replace VPW-P-00073 (Should bearing) if radial play exceeds 0.15 mm.

* 6000 Running Hours (PM-3):
- Execute all PM-1 and PM-2 tasks.
- Complete overhaul of the hot-air recirculation blower assembly.
- Full recalibration of all safety loops, emergency stop pushbuttons, and over-temperature limit thermostats.

5. FAULT-CODE TABLE AND CORRECTIVE ACTIONS

5.1 The control system logs specific fault codes for IND-OVEN irregularities. Technicians shall consult the following diagnostic matrix for probable causes and mandatory corrective actions:

1. Fault Code IND-FLT-01 (Vibration Warning / Trip):
* *Probable Cause:* Mechanical unbalance in the recirculation fan, loose motor mounting bolts, or failure of VPW-P-00073 (Should bearing).
* *Corrective Action:* Isolate electrical supply. Inspect blower impeller for material build-up or blade erosion. Measure bearing radial play; replace VPW-P-00073 (Should bearing) if worn. Retorque all motor base bolts.

2. Fault Code IND-FLT-02 (Surface Temperature High):
* *Probable Cause:* Degradation of internal thermal insulation blanket or failure of VPW-P-00058 (Reason seal) allowing hot-gas bypass.
* *Corrective Action:* Allow oven to cool below $40\,^{\circ}\text{C}$. Perform thermographic survey of casing exterior. Replace compromised sections of VPW-P-00058 (Reason seal) and re-pack internal insulation.

3. Fault Code IND-FLT-03 (Drive Motor Overcurrent):
* *Probable Cause:* Binding in the conveyor transport mechanism, excessive belt tension on VPW-P-00038 (Family belt) or VPW-P-00039 (Relationship belt), or seized transmission components.
* *Corrective Action:* Lock out electrical energy. Manually rotate conveyor drive train to check for mechanical binding. Inspect and readjust belt tracking and tension. Check motor winding resistance.

4. Fault Code IND-FLT-04 (Contactor Failure to Close / Chatter):
* *Probable Cause:* Low control circuit voltage, degraded contact surfaces on VPW-P-00048 (Part contactor) or VPW-P-00052 (Material contactor), or dust ingress in the magnetic coil housing.
* *Corrective Action:* Verify 24V DC control rail stability. Isolate power, open contactor housing, inspect silver-cadmium oxide contacts, and replace VPW-P-00048 or VPW-P-00052 if pitting exceeds 1.0 mm depth.

5. Fault Code IND-FLT-05 (Pneumatic Actuation Failure):
* *Probable Cause:* Internal spool leakage or contamination in VPW-P-00004 (Forget valve) or VPW-P-00069 (Race valve).
* *Corrective Action:* Isolate compressed air supply and bleed downstream pressure. Remove valve assemblies, flush with anhydrous solvent, inspect internal spools, and replace faulty valve units using exact plant inventory part numbers.

6. Fault Code IND-FLT-06 (Heating Element Control Loop Fault):
* *Probable Cause:* Solid-state relay failure, open-circuit heating element, or thermocouple drift.
* *Corrective Action:* Isolate 415V electrical supply. Measure element resistance across all three phases. Verify thermocouple millivolt output against temperature bath. Replace defective element banks or control modules.

6. CONSUMABLE AND SPARE PARTS LIST

6.1 Technicians shall only use genuine plant-inventories spare parts listed below for asset class IND-OVEN. Substitution of unlisted parts is strictly prohibited and invalidates safety certification.

Part Number	Plant Inventory Description	Standard Repl. Qty
VPW-P-00004	Forget valve	1 per assembly
VPW-P-00038	Family belt	2 per drive train
VPW-P-00039	Relationship belt	1 per drive train
VPW-P-00048	Part contactor	3 per main panel
VPW-P-00052	Material contactor	2 per main panel
VPW-P-00058	Reason seal	1 set per casing perimeter
VPW-P-00069	Race valve	2 per pneumatic manifold
VPW-P-00073	Should bearing	4 per conveyor shaft

7. TORQUE SPECIFICATIONS

7.1 Fasteners on asset class IND-OVEN shall be tightened using a calibrated torque wrench to the following limits:

* M6 Electrical Terminal Screws: 2.5 Nm ($\pm 0.2\text{ Nm}$)
* M8 Busbar Connections: 14.0 Nm ($\pm 1.0\text{ Nm}$)
* M10 Motor Base and Flange Bolts: 45.0 Nm ($\pm 3.0\text{ Nm}$)
* M12 Bearing Housing Retention Bolts: 78.0 Nm ($\pm 5.0\text{ Nm}$)

8. SAFETY-CRITICAL CORRECTIVE ACTIONS AND PERMIT REQUIREMENTS

8.1 The following corrective actions are designated as safety-critical due to the severe risk of electrical shock, arc flash, or uncontrolled thermal release:
1. Replacement or internal maintenance of power contactors VPW-P-00048 (Part contactor) and VPW-P-00052 (Material contactor).
2. Servicing, replacement, or rewiring of heating element banks and internal thermal insulation seals (VPW-P-00058).
3. Disassembly, alignment, or replacement of conveyor drive belts (VPW-P-00038, VPW-P-00039) and bearing assemblies (VPW-P-00073).
4. Intervention on pneumatic control valves VPW-P-00004 (Forget valve) and VPW-P-00069 (Race valve) when connected to pressurized headers.

8.2 Execution of any safety-critical corrective action listed in Clause 8.1 requires:
* An approved Lockout/Tagout (LOTO) application isolating Electrical (415V/24V) and Thermal energy sources.
* A mandatory Hot Work Permit if any welding, grinding, or thermal cutting is performed within 3.0 meters of the curing oven casing.
* A signed Confined Space Entry Permit prior to any technician entering the internal curing chamber.

End of Document — SOP-IND-OVEN-014